Nidharsan V – 22CSR130 III CSE C

Day 3 – Minikube installation and mysql

Kubernetes

Kubernetes (K8s) is an open-source container orchestration platform that automates the deployment, scaling, and management of containerized applications. It helps in efficiently managing multiple containers across a cluster of machines, ensuring high availability, load balancing, and self-healing capabilities. Kubernetes is widely used for cloud-native applications and microservices architectures.

Minikube

Minikube is a lightweight Kubernetes implementation that runs a single-node Kubernetes cluster on a local machine. It is primarily used for development and testing purposes, allowing developers to experiment with Kubernetes features without needing a full-scale cluster. Minikube supports various container runtimes and can be installed on Windows, macOS, and Linux

curl -LO https://github.com/kubernetes/minikube/releases/latest/download/minikube-linux-amd64

sudo install minikube-linux-amd64 /usr/local/bin/minikube && rm minikube-linux-amd64

minikube start

minikube start

minikube status

YML file

ersion: '3'

```
services:
  web:
    image: nginx:latest
    ports:
      - 80:80

  db:
    image: mysql:latest
    environment:
      - MYSQL_ROOT_PASSWORD=secret
```

```
docker exec -it david-db-1 /bin/bash
mysql -u root -p
```

Docker compose:

Docker Compose

Docker Compose is a tool that allows you to define and manage multi-container Docker applications using a YAML configuration file (docker-compose.yml). It simplifies the process of running multiple interdependent services (such as a web server, database, and caching system) with a single command.

Key Features:

- **Multi-Container Management** – Define multiple services in one file.
- **Service Dependencies** – Automatically starts services in the correct order.
- **Networking** – Easily creates a shared network for containers.
- **Scalability** – Scale services up or down with a single command.

Example docker-compose.yml:

yaml

Copy code

```yaml
version: '3'
services:
  web:
    image: nginx
    ports:
      - "8080:80"
  db:
    image: mysql
    environment:
      MYSQL_ROOT_PASSWORD: example
```

Usage:

sh

Copy code

```sh
# Start all services
docker compose up -d

# Stop and remove containers
docker compose down
```

Docker compose commands:

```sh
# Start and run containers in the background
docker compose up -d
```

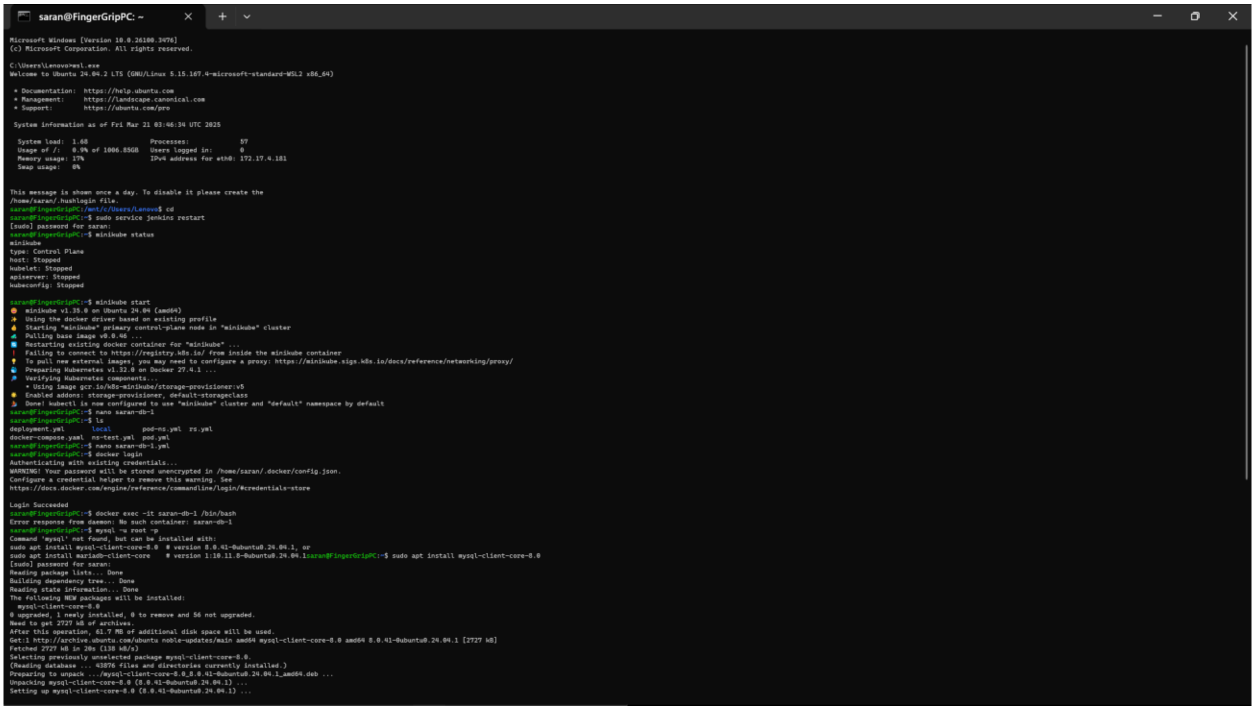
```
# Start containers in the foreground (logs will be shown)
docker compose up

# Stop containers
docker compose down

# Restart containers
docker compose restart

# View running containers
docker compose ps

# View logs of services
docker compose logs

# View logs of a specific service
docker compose logs <service_name>

# Build or rebuild services
docker compose build

# Stop containers without removing them
docker compose stop

# Start stopped containers
docker compose start
```

```
# Execute a command in a running container
docker compose exec <service_name> <command>

# Remove stopped containers, networks, and volumes
docker compose down --volumes

# Show configuration details
docker compose config

# Scale a service (e.g., run 3 instances of a service)
docker compose up --scale <service_name>=3 -d
```

Pipeline code

```
pipeline {
  agent any
  tools {maven "maven"}
  stages {
    stage('SCM') {
      steps {
        git branch: 'master', url: 'https://github.com/Saran-Avinash/DevOps.git'
      }
    }
    stage('Build') {
      steps {
        sh 'mvn clean package'
```

```
        }
      }
      stage('build to images') {
        steps {
          script {
            sh 'docker build -t saranavinashb/webapp1 .'
          }
        }
      }
      stage('push to hub') {
        steps {
          script {
            withDockerRegistry(credentialsId: 'docker_cred', toolName: 'docker', url:
'https://index.docker.io/v1/') {
                  sh 'docker push saranavinashb/webapp1'
            }
          }
        }
      }

  }
}
```



saran@FingerGripPC: ~

```
Microsoft Windows [Version 10.0.26100.3476]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Lenovo>wsl.exe
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 5.15.167.4-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Mar 21 03:46:34 UTC 2025

  System load:  1.68               Processes:             57
  Usage of /:   0.9% of 1006.85GB   Users logged in:       0
  Memory usage: 17%                IPv4 address for eth0: 172.17.4.181
  Swap usage:   0%


This message is shown once a day. To disable it please create the
/home/saran/.hushlogin file.
saran@FingerGripPC:/mnt/c/Users/Lenovo$ cd
saran@FingerGripPC:~$ sudo service jenkins restart
[sudo] password for saran:
saran@FingerGripPC:~$ minikube status
minikube
type: Control Plane
host: Stopped
kubelet: Stopped
apiserver: Stopped
kubeconfig: Stopped

saran@FingerGripPC:~$ minikube start
   minikube v1.35.0 on Ubuntu 24.04 (amd64)
   Using the docker driver based on existing profile
   Starting "minikube" primary control-plane node in "minikube" cluster
   Pulling base image v0.0.46 ...
   Restarting existing docker container for "minikube" ...
   Failing to connect to https://registry.k8s.io/ from inside the minikube container
   To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
   Preparing Kubernetes v1.32.0 on Docker 27.4.1 ...
   Verifying Kubernetes components...
    • Using image gcr.io/k8s-minikube/storage-provisioner:v5
   Enabled addons: storage-provisioner, default-storageclass
   Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
saran@FingerGripPC:~$ nano saran-db-1
saran@FingerGripPC:~$ ls
deployment.yml       local        pod-ns.yml  rs.yml
docker-compose.yaml  ns-test.yml  pod.yml
saran@FingerGripPC:~$ nano saran-db-1.yml
saran@FingerGripPC:~$ docker login
Authenticating with existing credentials...
WARNING! Your password will be stored unencrypted in /home/saran/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
saran@FingerGripPC:~$ docker exec -it saran-db-1 /bin/bash
Error response from daemon: No such container: saran-db-1
saran@FingerGripPC:~$ mysql -u root -p
Command 'mysql' not found, but can be installed with:
sudo apt install mysql-client-core-8.0  # version 8.0.41-0ubuntu0.24.04.1, or
sudo apt install mariadb-client-core    # version 1:10.11.8-0ubuntu0.24.04.1saran@FingerGripPC:~$ sudo apt install mysql-client-core-8.0
[sudo] password for saran:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  mysql-client-core-8.0
0 upgraded, 1 newly installed, 0 to remove and 56 not upgraded.
Need to get 2727 kB of archives.
After this operation, 61.7 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-client-core-8.0 amd64 8.0.41-0ubuntu0.24.04.1 [2727 kB]
Fetched 2727 kB in 20s (138 kB/s)
Selecting previously unselected package mysql-client-core-8.0.
(Reading database ... 43876 files and directories currently installed.)
Preparing to unpack .../mysql-client-core-8.0_8.0.41-0ubuntu0.24.04.1_amd64.deb ...
Unpacking mysql-client-core-8.0 (8.0.41-0ubuntu0.24.04.1) ...
Setting up mysql-client-core-8.0 (8.0.41-0ubuntu0.24.04.1) ...
```